Amal George

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Summary

Cybersecurity-focused Full Stack and Web3 Developer specializing in secure application design, smart contract auditing, end-to-end encryption, and MITRE ATT&CK-based threat detection. Engineered a SOC simulation platform with rule-based intrusion detection, an E2EE chat system using hybrid RSA/AES cryptography, and a decentralized Web3 dApp leveraging Solidity and Hardhat with custom on-chain governance. Combines production MERN development and EVM smart contract development with hands-on penetration testing (Kali Linux, Nmap, Metasploit).

Technical Skills

Languages: JavaScript, Python, C, C++
Frontend: HTML5, CSS3, React.js, Vite, Tailwind CSS
Backend: Node.js, Express.js, REST APIs, JWT Authentication, MVC Architecture
Databases: MongoDB, MySQL
Security: Application Security, Authentication & Authorization, End-to-End Encryption, AES Encryption, Public-Key Cryptography, OWASP, Vulnerability Assessment, Penetration Testing, MITRE ATT&CK Mapping
Networking: TCP/IP, HTTP/HTTPS, DNS, SSL/TLS, CORS
Operating Systems: Windows, Linux
Developer Tools: VS Code, Git, GitHub, Postman, Render, Vercel

Education

IIIT Kottayam **2023 – 2027**
BTech in Computer Science with Specialization in Cyber Security *CGPA : 7.03*

Vijayagiri Public School **2020 – 2022**
Higher Secondary Education *Percentage : 91.6*

Work Experience

Full Stack MERN Developer **Jul 2025 – Jan 2026**
Labmentix Private Limited

- Developed and maintained full-stack web applications using the MERN stack, delivering 10+ features and modules to improve application functionality, scalability, and user experience.
- Designed and integrated 20+ RESTful APIs, implemented JWT-based authentication, and optimized MongoDB queries, achieving a 10% improvement in API response time.
- Built responsive React.js interfaces, resolved 15+ production bugs and led code reviews across a Git-based team workflow, improving application stability and reducing regression incidents

Projects

CRYPTCHAT | *MERN Stack, Socket.IO, MongoDB, JWT, RSA-OAEP, AES-256-GCM* **May 2026**

- Built a security-focused real-time chat application using the MERN stack with Socket.IO, featuring End-to-End Encryption (E2EE) for confidential messaging.
- Executed hybrid encryption using RSA-OAEP, AES-256-GCM, and PBKDF2 to securely exchange keys, encrypt messages, and protect user's private keys on the client side.
- Integrated JWT authentication, HTTP-only cookies, Helmet, rate limiting, encrypted message storage, online user tracking, live notifications, and persistent chat history.

THREATLENS | *Python, React, FastAPI, Node.js, MongoDB, JWT* **June 2026**

- Designed and developed a Security Operations Center (SOC) simulation platform to ingest, parse, and analyze Linux authentication logs, web server logs, and firewall logs for threat detection and security event correlation.

- Engineered a rule-based threat detection engine to identify brute-force attacks, unauthorized access attempts, suspicious web requests, username enumeration, and anomalous network traffic, incorporating MITRE ATT&CK mapping and risk-based alert scoring.
- Implemented secure authentication, interactive security dashboards, alert management, and historical log analysis using React, FastAPI, Express.js, MongoDB, and Chart.js to support centralized security monitoring, threat investigation, and incident response.

FUNDCHAIN | *Blockchain, React, Solidity, Node.js, Hardhat, Sepolia, Metamask* **September 2026**

- Architected and deployed a decentralized Web3 crowdfunding dApp on the Sepolia testnet using Solidity, Hardhat, and Ethers.js v6, implementing a factory pattern to dynamically spin up isolated smart contracts.
- Engineered a trustless fund-governance model utilizing custom smart contract logic to lock raised capital and enforce majority-voting among contributors before executing spending proposals.
- Developed a responsive, dark-mode Web3 dashboard using React.js and Tailwind CSS, enabling real-time transaction state updates, wallet connection via MetaMask, and intuitive contract interactions.

Certifications

MERN Stack Web Development **Nov 2024 – Jun 2025**

Udemy – MasynTech

- Developed full-stack web applications using React, Node.js, Express.js, and MongoDB.
- Implemented RESTful APIs, JWT authentication, and secure database schemas across production-style projects.

Ethical Hacking & Security **Sept 2025 – Dec 2025**

Udemy – zSecurity

- Executed penetration testing, vulnerability assessments, and web/network security analysis.
- Utilized Kali Linux, Nmap, Wireshark, and Metasploit to identify and exploit security vulnerabilities.

Ethereum & Solidity Web3 Development **July 2026 – Aug 2026**

Udemy – Stephen Grider

- Engineered and deployed decentralized applications (dApps) using Solidity, Hardhat, Ethers.js, and React.
- Designed smart contract architectures implementing factory patterns, access controls, and custom governance logic.